

HIPAA Safe Harbor PHI/PII De-Identification Gateway

1.0.0 OAS 3.1

/openapi.json

Production-grade gateway service stripping patient identifiers before text reaches foundation LLMs and restoring clinical context on responses without leaking protected data.

System

GET

/health

Health Check

^

Health check, version metadata, and parameter verification.

Parameters

Cancel

No parameters

Execute

Clear

Responses

Curl

curl -X 'GET' \
 'http://localhost:8000/health' \
 -H 'accept: application/json'

Request URL

http://localhost:8000/health

Server response

Code	Details
200	Response body <pre>{ "status": "ok", "version": "1.0.0", "model_parameter_count": 124400000, "supported_adapters": ["mock", "openai", "anthropic", "gemini"] }</pre> Download Response headers <pre>content-length: 127 content-type: application/json date: Wed,02 Sep 2026 11:26:48 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response	No links

Media type

Code	Description	Links
	application/json Controls Accept header. Example Value Schema <pre>{ "status": "ok", "version": "1.0.0", "model_parameter_count": 0, "supported_adapters": ["string"] }</pre>	

De-Identification

POST

/deidentify

Deidentify Endpoint

De-identify clinical text according to HIPAA Safe Harbor 18 categories. Returns masked text with pseudonymised surrogate tokens and isolated mapping dict.

Parameters

Cancel

Reset

No parameters

Request body required

application/json

```
{
  "text": "PATIENT: Robert Henderson | DOB: 05/14/1958 | MRN: 884-9102-X | SSN: 078-45-9921\nDATE OF CONSULT: 10/14/2023\nATTENDING: Dr. James Parkinson, MD (NPI: 1982736450, Phone: 617-555-0144)\nFACILITY: St. Luke's Hospital, Boston, MA 02115\nREASON FOR CONSULT: Progressive resting tremor and cogwheel rigidity.\nHISTORY: Mr. Henderson is a 65-year-old male evaluated for Parkinson's disease. Prior appendectomy on 09/10/2023. Babinski reflex is negative bilaterally. Initiating carbidopa-levodopa 25/100 mg TID.\nPLAN: Follow-up clinic appointment on 11/25/2023 in 6 weeks.\nSigned: Dr. James Parkinson, MD",
  "patient_id": "PAT_8849102",
  "date_shift_days": -42,
  "masking_mode": "surrogate_token",
  "preserve_eponyms": true,
  "preserve_relative_dates": true
}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8000/deidentify' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "text": "PATIENT: Robert Henderson | DOB: 05/14/1958 | MRN: 884-9102-X | SSN: 078-45-9921\nDATE OF CONSULT: 10/14/2023\nATTENDING: Dr. James Parkinson, MD (NPI: 1982736450, Phone: 617-555-0144)\nFACILITY: St. Luke's Hospital, Boston, MA 02115\nREASON FOR CONSULT: Progressive resting tremor and cogwheel rigidity.\nHISTORY: Mr. Henderson is a 65-year-old male evaluated for Parkinson's disease. Prior appendectomy on 09/10/2023. Babinski reflex is negative bilaterally. Initiating carbidopa-levodopa 25/100 mg TID.\nPLAN: Follow-up clinic appointment on 11/25/2023 in 6 weeks.\nSigned: Dr. James Parkinson, MD",
    "patient_id": "PAT_8849102",
    "date_shift_days": -42,
    "masking_mode": "surrogate_token",
    "preserve_eponyms": true,
    "preserve_relative_dates": true
  }'
```

Request URL

http://localhost:8000/deidentify

Server response

Code	Details
200	Response body <pre>"shifted_str": "04/02/1958", "surrogate_token": "[DATE_1]", "delta_days": -42 }, { "original_str": "10/14/2023", "shifted_str": "09/02/2023", "surrogate_token": "[DATE_2]", "delta_days": -42 }, { "original_str": "09/10/2023", "shifted_str": "07/30/2023", "surrogate_token": "[DATE_3]", "delta_days": -42 }, { "original_str": "11/25/2023", "shifted_str": "10/14/2023", "surrogate_token": "[DATE_4]", "delta_days": -42 }], "bracket_map": {} }, "entity_count": 14, "date_shift_days": -42 }</pre> Response headers <pre>access-control-allow-credentials: true access-control-allow-origin: * content-length: 4142 content-type: application/json date: Mon,31 Aug 2026 18:52:24 GMT server: uvicorn</pre>

Download

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema <pre>{ "masked_text": "string", "mapping": {}, "entity_count": 0, "date_shift_days": 0 }</pre> No links	
422	Validation Error Media type application/json Example Value Schema <pre>{ "detail": [{ "loc": ["string", 0], "msg": "string", "type": "string" }] }</pre> No links	



POST /rehydrate Rehydrate Endpoint



Rehydrate foundation LLM response by substituting surrogate tokens with original entities.

Parameters

Cancel

Reset

No parameters

Request body required

application/json

```
{
  "response": "### Clinical Assessment & Plan\n* **Patient Name:** [PATIENT_1] (DOB: [DATE_1], MRN: [MRN_1])\n* **Consult Date:** [DATE_2]\n* **Attending Neurologist:** [PROVIDER_1] (NPI: [NPI_1], Phone: [PHONE_1])\n* **Facility:** [HOSPITAL_1], MA [ZIP_1]\n* **Diagnosis:** Parkinson's disease under active evaluation.\n* **Medication Plan:** Initiated carbidopa-levodopa 25/100 mg TID.\n* **Follow-up:** Scheduled for [DATE_4].",
  "mapping": {
    "token_to_original": {
      "[PATIENT_1]": "Robert Henderson",
      "[DATE_1]": "05/14/1958",
      "[MRN_1]": "884-9102-X",
      "[SSN_1]": "078-45-9921",
      "[DATE_2]": "10/14/2023",
      "[PROVIDER_1]": "Dr. James Parkinson, MD",
      "[NPI_1]": "1982736450",
      "[PHONE_1]": "617-555-0144",
      "[HOSPITAL_1]": "St. Luke's Hospital, Boston",
      "[ZIP_1]": "02115",
      "[PATIENT_2]": "Henderson",
      "[DATE_3]": "09/10/2023",
      "[DATE_4]": "11/25/2023"
    }
  }
}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8000/rehydrate' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "response": "### Clinical Assessment & Plan\n* **Patient Name:** [PATIENT_1] (DOB: [DATE_1], MRN: [MRN_1])\n* **Consult Date:** [DATE_2]\n* **Attending Neurologist:** [PROVIDER_1] (NPI: [NPI_1], Phone: [PHONE_1])\n* **Facility:** [HOSPITAL_1], MA [ZIP_1]\n* **Diagnosis:** Parkinson's disease under active evaluation.\n* **Medication Plan:** Initiated carbidopa-levodopa 25/100 mg TID.\n* **Follow-up:** Scheduled for [DATE_4].",
    "mapping": {
      "token_to_original": {
        "[PATIENT_1]": "Robert Henderson",
        "[DATE_1]": "05/14/1958",
        "[MRN_1]": "884-9102-X",
        "[SSN_1]": "078-45-9921",
        "[DATE_2]": "10/14/2023",
        "[PROVIDER_1]": "Dr. James Parkinson, MD",
        "[NPI_1]": "1982736450",
        "[PHONE_1]": "617-555-0144",
        "[HOSPITAL_1]": "St. Luke's Hospital, Boston",
        "[ZIP_1]": "02115",
        "[PATIENT_2]": "Henderson",
        "[DATE_3]": "09/10/2023",
        "[DATE_4]": "11/25/2023"
      },
      "date_mappings": [],
      "bracket_map": {}
    }
  },
  "strict_mode": false
}
```

Request URL

http://localhost:8000/rehydrate

Server response

Code	Details
200	Response body <pre>{ "rehydrated_text": "### Clinical Assessment & Plan\n* **Patient Name:** Robert Henderson (DOB: 05/14/1958, MRN: 884-9102-X)\n* **Consult Date:** 10/14/2023\n* **Attending Neurologist:** Dr. James Parkinson, MD (NPI: 1982736450, Phone: 617-555-0144)\n* **Facility:** St. Luke's Hospital, Boston, MA 02115\n* **Diagnosis:** Parkinson's disease under active evaluation.\n* **Medication Plan:** Initiated carbidopa-levodopa 25/100 mg TID.\n* **Follow-up:** Scheduled for 11/25/2023.", "restored_entities_count": 13 }</pre> Download Response headers <pre>access-control-allow-credentials: true access-control-allow-origin: * content-length: 509 content-type: application/json date: Mon,31 Aug 2026 18:54:28 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema <pre>{ "rehydrated_text": "string", "restored_entities_count": 0 }</pre>	No links
422	Validation Error Media type application/json Example Value Schema <pre>{ "detail": [{ "loc": ["string", 0], "msg": "string", "type": "string" }] }</pre>	No links

Gateway

POST /gateway/process Gateway Process Endpoint

Full end-to-end roundtrip: Raw Note -> De-identify -> LLM Generation -> Rehydrate.

Parameters

Cancel

Reset

No parameters

Request body required

application/json

```
{
  "clinical_note": "ADMISSION NOTE & OPERATIVE PLAN\nPATIENT: Eleanor Vance (MRN: 48201, DOB: 10/14/1982)\nAttending Surgeon: Dr. Alan Whipple, MD at Massachusetts General Hospital, Boston, MA 02114\nAssistant Surgeon: Dr. Katherine Vance, MD\nDiagnosis: Whipple disease and pancreatic head adenocarcinoma.\nProcedure: Classic pancreaticoduodenectomy (Whipple procedure) scheduled.\nPlan: Prescribed Ceftriaxone IV for 2 weeks followed by oral TMP-SMX for 1 year.",
  "task_prompt": "Please generate a concise structured clinical summary, surgical evaluation, and antibiotic regimen for this patient note:\n\n{text}",
  "adapter_provider": "gemini",
  "model_name": "gemini-3.6-flash",
  "patient_id": "PAT_48201",
  "date_shift_days": -42
}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8000/gateway/process' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "clinical_note": "ADMISSION NOTE & OPERATIVE PLAN\nPATIENT: Eleanor Vance (MRN: 48201, DOB: 10/14/1982)\nAttending Surgeon: Dr. Alan Whipple, MD at Massachusetts General Hospital, Boston, MA 02114\nAssistant Surgeon: Dr. Katherine Vance, MD\nDiagnosis: Whipple disease and pancreatic head adenocarcinoma.\nProcedure: Classic pancreaticoduodenectomy (Whipple procedure) scheduled.\nPlan: Prescribed Ceftriaxone IV for 2 weeks followed by oral TMP-SMX for 1 year.",
    "task_prompt": "Please generate a concise structured clinical summary, surgical evaluation, and antibiotic regimen for this patient note:\n\n{text}",
    "adapter_provider": "gemini",
    "model_name": "gemini-3.6-flash",
    "patient_id": "PAT_48201",
    "date_shift_days": -42
  }'
```

Request URL

http://localhost:8000/gateway/process

Server response

Code	Details
200	Response body <pre>{ "final_text": "Here is a concise structured clinical summary, surgical evaluation, and antibiotic regimen based on the provided note:\n\n---\n\n### **1. Clinical Summary**\n\n**Patient Name:** Eleanor Vance\n**MRN:** 48201\n**DOB:** 10/14/1982\n**Attending Surgeon:** Dr. Alan Whipple, MD at Massachusetts General Hospital (Boston, MA)\n**Assistant Surgeon:** Dr. Katherine Vance, MD\n**Diagnoses:** \n 1. Pancreatic head adenocarcinoma\n 2. Whipple disease (*Tropheryma whipplei* infection)\n\n---\n\n### **2. Surgical Evaluation & Plan**\n\n**Indication:** Pancreatic head adenocarcinoma\n**Planned Procedure:** Classic Pancreaticoduodenectomy (Whipple Procedure)\n**Status:** Scheduled\n\n---\n\n### **3. Antibiotic Regimen**\n\n**Indication:** Medical management of Whipple disease\n**Initial Therapy:** Ceftriaxone IV daily for 2 weeks\n**Maintenance Therapy:** Oral Trimethoprim-Sulfamethoxazole (TMP-SMX) for 1 year post-initial therapy", "masked_input": "ADMISSION NOTE & OPERATIVE PLAN\nPATIENT: [PATIENT_1] (MRN: [MRN_1], DOB: [DATE_1])\nAttending Surgeon: [HOSPITAL_1], [CITY_1], MA [ZIP_1]\nAssistant Surgeon: [PROVIDER_1]\nDiagnosis: Whipple disease and pancreatic head adenocarcinoma.\nProcedure: Classic pancreaticoduodenectomy (Whipple procedure) scheduled.\nPlan: Prescribed Ceftriaxone IV for 2 weeks followed by oral TMP-SMX for 1 year.", "raw_llm_response": "Here is a concise structured clinical summary, surgical evaluation, and antibiotic regimen based on the provided note:\n\n---\n\n### **1. Clinical Summary**\n\n**Patient Name:** [PATIENT_1]\n**MRN:** [MRN_1]\n**DOB:** [DATE_1]\n**Attending Surgeon:** [HOSPITAL_1] ([CITY_1], MA)\n**Assistant Surgeon:** [PROVIDER_1]\n**Diagnoses:** \n 1. Pancreatic head adenocarcinoma\n 2. Whipple disease (*Tropheryma whipplei* infection)\n\n---\n\n### **2. Surgical Evaluation & Plan**\n\n**Indication:** Pancreatic head adenocarcinoma\n**Planned Procedure:** Classic Pancreaticoduodenectomy (Whipple Procedure)\n**Status:** Scheduled\n\n---\n\n### **3. Antibiotic Regimen**\n\n**Indication:** Medical management of Whipple disease\n**Initial Therapy:** Ceftriaxone IV daily for 2 weeks\n**Maintenance Therapy:** Oral Trimethoprim-Sulfamethoxazole (TMP-SMX) for 1 year post-initial therapy", "mapping": { "version": "1.0.0", "document_id": "doc_20599b49-a897-4e18-b56c-25def21ea907", "created_at": "2026-08-31T18:57:58.303345+00:00", "date_shift_days": -42, "entities": [</pre> Download Response headers <pre>access-control-allow-credentials: true access-control-allow-origin: *</pre>

Code	Details	
	content-length: 4387 content-type: application/json date: Mon,31 Aug 2026 18:57:56 GMT server: uvicorn	
Responses		
Code	Description	Links
200	Successful Response	No links
Media type application/json Controls Accept header. Example Value Schema <pre>{ "final_text": "string", "masked_input": "string", "raw_llm_response": "string", "mapping": {}, "latency_ms": 0, "deid_latency_ms": 0, "llm_latency_ms": 0, "rehydrate_latency_ms": 0, "entity_count": 0, "leak_check_passed": true }</pre>		
422	Validation Error	No links
Media type application/json Example Value Schema <pre>{ "detail": [{ "loc": ["string", 0], "msg": "string", "type": "string" }] }</pre>		

POST

/gateway/summarize

Gateway Summarize Endpoint

^

Convenience endpoint for clinical summarization through the secure gateway.

Parameters

Cancel

Reset

No parameters

Request body required

application/json

```
{
  "clinical_note": "GERIATRIC INTAKE NOTE\nPATIENT: Arthur Pendelton | AGE: 94-year-old male | DOB: 01/18/1930\nMRN: GER-392019 | DATE: 04/10/2024\nATTENDING: Dr. Rebecca Sterling, MD (Phone: 312-555-0199)\nFACILITY: Oakwood Extended Care, Chicago, IL 60611\nASSESSMENT: 94yo nonagenarian admitted for mild cognitive impairment. Celebrated his 90th birthday in 2020. Patient ambulates with a walker.\nPLAN: Continue physical therapy 3 times weekly. Follow-up on 05/10/2024.",
  "adapter_provider": "gemini",
  "model_name": "gemini-3.6-flash"
}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8000/gateway/summarize' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "clinical_note": "GERIATRIC INTAKE NOTE\nPATIENT: Arthur Pendelton | AGE: 94-year-old male | DOB: 01/18/1930\nMRN: GER-392019 | DATE: 04/10/2024\nATTENDING: Dr. Rebecca Sterling, MD (Phone: 312-555-0199)\nFACILITY: Oakwood Extended Care, Chicago, IL 60611\nASSESSMENT: 94yo nonagenarian admitted for mild cognitive impairment. Celebrated his 90th birthday in 2020. Patient ambulates with a walker.\nPLAN: Continue physical therapy 3 times weekly. Follow-up on 05/10/2024.",
    "adapter_provider": "gemini",
    "model_name": "gemini-3.6-flash"
  }'
```

Request URL

http://localhost:8000/gateway/summarize

Server response

Code	Details
200	Response body <pre>{ "original_str": "01/18/1930", "shifted_str": "06/15/1930", "surrogate_token": "[DATE_1]", "delta_days": 148 }, { "original_str": "04/10/2024", "shifted_str": "09/05/2024", "surrogate_token": "[DATE_2]", "delta_days": 148 }, { "original_str": "05/10/2024", "shifted_str": "10/05/2024", "surrogate_token": "[DATE_3]", "delta_days": 148 }], "bracket_map": {} }, "latency_ms": 7190.325499999744, "deid_latency_ms": 228.7797999997565, "llm_latency_ms": 6961.3104000001821, "rehydrate_latency_ms": 0.212800000146362, "entity_count": 13, "leak_check_passed": true }</pre> Download Response headers <pre>access-control-allow-credentials: true access-control-allow-origin: * content-length: 4852 content-type: application/json date: Mon,31 Aug 2026 19:03:56 GMT</pre>

Code	Details
	server: uvicorn
Responses	
Code	DescriptionLinks
200	Successful ResponseNo links
	Media type application/json Controls Accept header. Example ValueSchema <pre>{ "final_text": "string", "masked_input": "string", "raw_llm_response": "string", "mapping": {}, "latency_ms": 0, "deid_latency_ms": 0, "llm_latency_ms": 0, "rehydrate_latency_ms": 0, "entity_count": 0, "leak_check_passed": true}</pre>
422	Validation ErrorNo links
	Media type application/json Example ValueSchema <pre>{ "detail": [{ "loc": ["string", 0], "msg": "string", "type": "string" }]}</pre>

Schemas



- DeidentifyRequest > Expand all object
- DeidentifyResponse > Expand all object
- GatewayProcessRequest > Expand all object
- GatewayProcessResponse > Expand all object
- GatewaySummarizeRequest > Expand all object
- HTTPValidationError > Expand all object
- HealthResponse > Expand all object

RehydrateRequest > Expand all **object**

RehydrateResponse > Expand all **object**

ValidationError > Expand all **object**